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Reg No.:_____ Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Eighth Semester B.Tech Degree (R,S) Examination May 2024 (2019 Scheme)

Course Code: CST466

Course Name: DATA MINING

Max. Marks: 100 **Duration: 3 Hours**

PART A

Answer all questions, each carries 3 marks. Marks

1 Illustrate the multi-dimensional data model with a neat figure. (3)

2 List the differences between OLAP and OLTP. (3)

3 Justify the significance of pre-processing the data before mining. (3)

4 Explain the two sampling methods used in data reduction. (3)

5 Given two objects are represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8). Compute (i) Euclidean distance ii) Manhattan distance (iii) Minkowski distance of order 3. (3)

6 Draw the confusion matrix and calculate precision and recall of the given data. (3)

Data	Target	Prediction
1	cat	cat
2	cat	dog
3	dog	dog
4	dog	dog
5	dog	cat

7 Discuss the significance of association rule mining in market basket analysis. (3)

8 List the modification methods to improve the efficiency of Apriori algorithm. (3)

9 Compare web structure mining and web usage mining. (3)

10 Distinguish between focused crawling and regular crawling. (3)

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

11 a) Explain the three-tier architecture of the data warehouse with a neat figure. (6)

b) List and illustrate the schemas used for the physical representation of the multi-dimensional data with examples. (8)

OR

12 a) List and explain various data mining functionalities. (6)

b) Illustrate different OLAP operations in multidimensional data model with examples. List the differences between ROLAP, MOLAP and HOLAP. (8)

Module II

13 a) Discuss the significance of data discretization in data mining. List and explain any four data discretization strategies. (6)

b) Illustrate PCA for dimensionality reduction with an example. (8)

OR

14 a) Explain various methods for dealing with missing data. Justify its significance. (6)

b) Explain data normalization methods with necessary equations. Calculate the normalized value for 550 in the set of the data points 50, 150, 250, 350, 450, 550, 650, 700, 850, 950, 1000 using (8)

(a) min-max normalization by setting min = 0 and max = 1

(b) z-score normalization

(c) Normalization by decimal scaling

Module III

15 a) Illustrate the clustering technique using DBSCAN. (6)

b) Explain the steps in creating decision tree using ID3 algorithm. For the dataset given below, find the first splitting attribute for the decision tree by using the ID3 algorithm. (8)

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

OR

16 a) Discuss PAM algorithm for clustering with an example. (6)

b) Explain the construction of decision tree using SLIQ algorithm with an example. (8)

Module IV

17 a) Illustrate the working of Pincer Search algorithm with an example. (6)

b) State the Apriori principle in candidate generation. Find out the frequent item sets with minimum support of 2 using Apriori for the following data. (8)

TID	ITEMSETS
T1	A, B
T2	B, D
T3	B, C
T4	A, B, D
T5	A, C
T6	B, C
T7	A, C
T8	A, B, C, E
T9	A, B, C

OR

18 a) Discuss the partitioning algorithm for finding large itemset and compare its performance with Apriori algorithm. (6)

b) Mention the advantages of FP Growth algorithm. Find out the frequent item sets using FP Growth for the following data. (8)

TID	ITEMSETS
T1	A, B
T2	B, D
T3	B, C
T4	A, B, D
T5	A, C
T6	B, C
T7	A, C
T8	A, B, C, E
T9	A, B, C

Module V

19 a) Describe web content mining techniques. (6)

b) Discuss various text mining approaches and techniques in detail. (8)

OR

20 a) Describe different Text retrieval methods. Distinguish text mining, information retrieval and information extraction. (6)

b) Write and explain CLEVER algorithm for web structure mining. (8)

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